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Lic. in Informatics and Computing Engineering (L.EIC)

Discrete Mathematics

Extra 4 - Semantics of First Order Logic

4.1. Consider the language, $\mathcal{L}$, a language of first order with equality and such that $\mathcal{F}_0 = \{a, b\}$, $\mathcal{F}_1 = \{g\}$, $\mathcal{F}_2 = \{f, h\}$, $\mathcal{R}_1 = \{R, T\}$ and $\mathcal{R}_2 = \{P, Q\}$. For each of the following propositions, give a structure of $\mathcal{L}$ where it is true and another where it is false. Can you conclude that neither the propositions nor their negations are valid formulas?

(a) $\forall x\,(x = a)$
(b) $\forall x \forall y\,(R(x) \to R(y))$
(c) $\exists z\,((g(z) \neq a) \to \exists z\,(g(z) = a))\ \forall x\,Q(x, g(x)) \leftrightarrow \exists x\,P(a, x)$
(d) $\forall y(\forall x\,(R(x) \to P(x, x)) \to (R(y) \to \forall x\,P(x, x)))$
(e) $\forall x \exists y\,(x = g(y))$
(f) $\exists y \forall x\,R(x, y) \to \forall x \exists y\,R(x, y)$
(g) $\exists x \forall y\,(x = g(y))$
(h) $\forall y \exists x\,R(x, y) \to \exists x \forall y\,R(x, y)$
(i) $\exists x\,(h(x) = g(a) \to \exists z\,(g(z) = a))$
(j) $\forall x\,Q(x, g(x)) \to \exists x\,R(a, x)$
(k) $\exists z\,((g(z) \neq a) \to \exists z\,(g(z) = a))$
(l) $\exists x\,Q(x, g(x)) \to \forall x\,R(a, x)$
(m) $\forall y \exists x\,R(x, y)$
(n) $\forall x\,(T(x) \to \exists y\,Q(x, y))$
(o) $\exists y \exists x\,(g(x) \neq g(y)$
(p) $\exists x\,(T(x) \to \forall y\,Q(x, y))$
(q) $\forall y \exists x\,(x \neq g(y))$
(r) $\forall x\,(T(x) \to \forall y\,Q(x, y))$
(s) $\forall y \exists x\,Q(x, g(y))$
(t) $\forall x \exists y\,(Q(x, y) \to T(y))$

4.2. Write each of the following propositions so that negations only appear on predicates (i.e. no negation is over a quantifier or an expression involving connectives).

(a) $\neg\exists y\, \exists x\, P(x, y)$

(b) $\neg\forall x\, \exists y\, P(x, y)$

(c) $\neg\exists y\, (Q(y) \wedge \forall x\, \neg R(x, y))$

(d) $\neg\exists y\, (\exists x\, R(x, y) \vee \forall x\, S(x, y))$

(e) $\neg\exists y\, (\forall x\, \exists z\, T(x, y, z) \vee \exists x\, \forall z\, U(x, y, z))$

4.3. For each of the following formulas determine an equivalent prenex normal form.

(a) $\neg(\exists x\, Q(x, y) \to \neg\forall x\, (\forall z P(z, y, x) \to \neg\exists y\, P(y)))$

(b) $\neg(\forall x\, R(x, y) \to \exists x\, \neg(\exists z\, Q(z, y, x) \to \forall y\, P(y)))$

(c) $\neg(\forall x\, Q(x) \vee \neg\forall x\, (\exists z\, R(z, x) \to \neg\forall y\, R(y, x)))$

Laws of semantic equivalence

$\phi \wedge \psi$	$\Leftrightarrow$	$\psi \wedge \phi$	(commutativity)
$\phi \vee \psi$	$\Leftrightarrow$	$\psi \vee \phi$	(commutativity)
$\neg(\phi \wedge \psi)$	$\Leftrightarrow$	$(\neg\phi \vee \neg\psi)$	(DeMorgan's Law)
$\neg(\phi \vee \psi)$	$\Leftrightarrow$	$(\neg\phi \wedge \neg\psi)$	(DeMorgan's Law)
$(\phi \wedge \psi) \wedge \theta$	$\Leftrightarrow$	$\phi \wedge (\psi \wedge \theta)$	(associativity)
$(\phi \vee \psi) \vee \theta$	$\Leftrightarrow$	$\phi \vee (\psi \vee \theta)$	(associativity)
$(\phi \vee \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \phi)$	$\Leftrightarrow$	ϕ	(idempotence)
$(\phi \wedge \psi) \vee \theta$	$\Leftrightarrow$	$(\phi \vee \theta) \wedge (\psi \vee \theta)$	(distributiveness)
$(\phi \vee \psi) \wedge \theta$	$\Leftrightarrow$	$(\phi \wedge \theta) \vee (\psi \wedge \theta)$	(distributivity)
$\neg\neg\phi$	$\Leftrightarrow$	ϕ	(Double negation)
$\phi \to \psi$	$\Leftrightarrow$	$\neg\phi \vee \psi$	
$\neg\forall x\varphi$	$\Leftrightarrow$	$\exists x\neg\varphi$	(DeMorgan's Law)
$\neg\exists x\varphi$	$\Leftrightarrow$	$\forall x\neg\varphi$	(DeMorgan's Law)
$\forall x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\forall x\varphi \wedge \forall x\psi$	(but not with $\vee$)
$\exists x\varphi \vee \exists x\psi$	$\Leftrightarrow$	$\exists x(\varphi \vee \psi)$	(but not with $\wedge$)
$\forall x\forall y\varphi$	$\Leftrightarrow$	$\forall y\forall x\varphi$	
$\exists x\exists y\varphi$	$\Leftrightarrow$	$\exists y\exists x\varphi$	
$\forall x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\forall x\varphi \wedge \psi$	if x does not occur free in ψ
$\forall x(\varphi \vee \psi)$	$\Leftrightarrow$	$\forall x\varphi \vee \psi$	if x does not occur free in ψ
$\exists x(\varphi \wedge \psi)$	$\Leftrightarrow$	$\exists x\varphi \wedge \psi$	if x does not occur free in ψ
$\exists x(\varphi \vee \psi)$	$\Leftrightarrow$	$\exists x\varphi \vee \psi$	if x does not occur free in ψ